Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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1. **Consider two diagonal matrices of order n:**

$$A = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix}, \qquad B = \begin{bmatrix} b_1 & 0 & \cdots & 0 \\ 0 & b_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_n \end{bmatrix}.$$

   a) **Calculate the matrix AB.**

   By the definition of matrix multiplication,

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

   Since A and B are diagonal matrices, all terms vanish except when $i = j = k$. Therefore,

$$\boxed{AB = \begin{bmatrix} a_1 b_1 & 0 & \cdots & 0 \\ 0 & a_2 b_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n b_n \end{bmatrix}}.$$

   Thus, the product of two diagonal matrices is also diagonal.

   b) **Determine the conditions that $a_1, a_2, \ldots, a_n$ must satisfy for A to be invertible. When A is invertible, determine its inverse.**

   We prove that

$$A \text{ is invertible} \iff \forall i \in \{1, \ldots, n\},\ a_i \neq 0.$$

   Necessity ($\Rightarrow$).

   Assume that A is invertible.

   If there exists $i \in \{1, \ldots, n\}$ such that $a_i = 0$, then the i-th row of A is zero.

   Consequently, the i-th row of AX is zero for every matrix $X \in M_n(K)$.

In particular, $AX \neq I_n$, contradicting the invertibility of A.
Therefore,

$$\forall i \in \{1, \ldots, n\},\ a_i \neq 0.$$

Sufficiency ($\Leftarrow$).
Assume that

$$\forall i \in \{1, \ldots, n\},\ a_i \neq 0.$$

Consider the diagonal matrix

$$X = \begin{bmatrix} a_1^{-1} & 0 & \cdots & 0 \\ 0 & a_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n^{-1} \end{bmatrix}.$$

By part a), both products AX and XA are diagonal, with diagonal entries

$$a_i a_i^{-1} = a_i^{-1} a_i = 1.$$

Hence,

$$AX = XA = I_n.$$

Therefore, A is invertible and $X = A^{-1}$.

Conclusion.

$$\boxed{A \text{ is invertible } \iff \forall i \in \{1, \ldots, n\},\ a_i \neq 0}$$

and, whenever A is invertible,

$$\boxed{A^{-1} = \begin{bmatrix} \frac{1}{a_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{a_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{a_n} \end{bmatrix}}.$$